

Tide retrospective:

Bounded-prior quartic partition function (grammar precursor)

Daniel Murfet (with Claude Opus 4.7)

2 May 2026

Abstract

A single Tide-loop excursion in the `laplace` seabed, formalising the smallest possible step toward the grammar paper’s $Z_n[\varphi] = \int \varphi(w) e^{-nK(w)} \pi(w) dw$ machinery: replace the integration domain $\mathbb{R}$ in the existing `quartic_partition` by a bounded interval $[-a, a]$ and quantify the resulting boundary correction. The headline theorem `Laplace.OneD.QuarticBoundedPrior.quartic_partition.bound` adds 232 lines of Lean to the seabed. Zero `sorry`, zero `axiom`, zero `native_decide`; clean diagnostics on first build after a parser fix on the absolute-value statement. The interesting moves were upstream: a deliberation step that took an over-ambitious direction (the full 2D $K = w_1^2 w_2^4$ resolution example, RLCT 3/4 with logarithm) and pushed it down to a 1D one-ingredient-at-a-time precursor, and a numerical sanity check that surfaced a loose exponent constant in the candidate statement before any Lean was written.

1 Setting

This Tide step sits in the same `laplace` seabed used by the preceding tide steps (`tide/quartic-moments`, `tide/2d-semi-degenerate`, `tide/universal-scaling`, `tide/f1-large-t`, `tide/2d-pure-quartic`), but it is the first tide aimed not at the Susceptibility Primer’s Section 4 but at the *grammar paper* (Gerraty et al., *Expectations and the Exceptional Divisor*). Per the cross-tide candidates survey (`projects/automation/log/2026-05-02-tide-candidates-survey.md`), the grammar paper’s *simplest resolution example* — $K(w) = w_1^2 \cdot w_2^4$ on a 2D parameter space with bounded prior, leading asymptotic $Z_n[\varphi] \sim C_\varphi \cdot n^{-3/4} \log n$ with RLCT $\lambda = 3/4$ and multiplicity $m = 2$ — is the natural next target, but the survey itself flagged the full 2D form as “probably the wrong shape for a single excursion” and asked Step 2 to pick a 1D precursor instead.

The Step 2 deliberation drafted three candidates of varying ambition: (A) bounded-prior quartic partition function with explicit exponential remainder; (B) the same with a harmonic potential (essentially glue-of-existing-seabed-lemmas); (C) bounded-prior quartic with a continuous test function (the genuine 1D Watanabe asymptotic at a regular critical point). GPT-5.5 Pro and Claude both backed A, on matched grounds: B is too small to count as a step outward (the harmonic tail is already in the seabed and there is no new quartic-side analysis to do), C is too large because it bundles bounded-prior, test-function localisation, change-of-variables, and dominated convergence in one step. A introduces *exactly one* ingredient (bounded prior, with the necessary new quartic tail bound) on the most-recently-finished tide step’s potential. The full deliberation is in `projects/grammar/tide-log/2026-05-02-tide-grammar-precursor.md`; the GPT consult is preserved verbatim alongside as `gpt55_tide_grammar_v1.md`.

The branch (`tide/grammar-precursor-bounded-prior`) is chained on top of `tide/2d-pure-quartic`’s tip (`efafbfa`) per the linear-chain branch-hygiene rule, so a single SHA-pin covers all in-flight tides on the way to a squash-merge.

2 The thing formalised

For the centred quartic potential

$$L(w) = \frac{w^4}{24}$$

and any $a > 0$, $t > 0$, the bounded-prior partition function differs from the existing full-line `quartic_partition` by an exponentially small correction:

$$\left| \int_{-a}^a e^{-t \cdot w^4 / 24} dw - \frac{1}{2} (24/t)^{1/4} \Gamma(1/4) \right| \leq \frac{24}{t \cdot a^3} \exp\left(-\frac{t \cdot a^4}{24}\right).$$

The Lean signature:

```
theorem quartic_partition_bounded_prior {t a : ℝ} (ht : 0 < t) (ha : 0 < a) :
  | ( w in Icc (-a) a, Real.exp (-(t * w ^ 4 / 24))) -
    (1/2) * (24/t) ^ ((1 : ℝ) / 4) * Real.Gamma ((1 : ℝ) / 4) | ≤
    (24 / (t * a ^ 3)) * Real.exp (-(t * a ^ 4 / 24))
```

located at `Laplace/OneD/QuarticBoundedPrior.lean`, in namespace `Laplace.OneD`.

This is the simplest possible step toward the grammar paper's $Z_n[\varphi]$ machinery. The grammar paper's partition function differs from the existing seabed in three ways: (i) bounded prior π replacing the full real line, (ii) test function φ inside the integrand, (iii) explicit asymptotic in n with remainder. The present theorem adds *only* (i), on the most degenerate 1D potential we already have, with a remainder that is not just asymptotic but *exact* for all $t > 0$. Subsequent tide steps can layer (ii) and the monomial generalisation on top, with the bounded-prior tail-bound infrastructure already in place.

Numerical sanity check: `scipy.integrate.quad` on $[-a, a]$ for $(a, t) \in \{1, 2\} \times \{10, 100, 1000\}$ agreed with leading + tail bound to machine precision (e.g. at $a = 1, t = 100$: numeric 1.2672, leading 1.2688, error 1.6×10^{-3} , tail bound 3.7×10^{-3}). The same script also validated candidate B (harmonic) and candidate C (cosine test function, ratio $\rightarrow 1$ slowly as $t \rightarrow \infty$); recorded in the tide log.

3 Proof strategy

The proof has three pieces: a tail sublemma, a tail-symmetry lemma, and a splitting identity that reduces the headline to combining them.

The *key sublemma* is `quartic_tail_Ioi`:

$$\int_a^\infty e^{-t \cdot w^4 / 24} dw \leq \frac{12}{t \cdot a^3} \exp\left(-\frac{t \cdot a^4}{24}\right).$$

The proof reduces the quartic tail to the existing Gaussian tail. For $w \geq a$ we have $w^2 \geq a^2$, hence $w^4 = w^2 \cdot w^2 \geq a^2 \cdot w^2$, hence

$$e^{-(t/24)w^4} \leq e^{-(ta^2/24)w^2} = e^{-(b/2)w^2} \quad \text{with } b = ta^2/12.$$

Apply the seabed's `Laplace.OneD.gaussian_tail_bound_rescaled_Ioi` at $M = a$, $b = ta^2/12$:

$$\int_a^\infty e^{-(bw^2)/2} dw \leq \frac{e^{-ba^2/2}}{b \cdot a} = \frac{12}{ta^3} \exp\left(-\frac{ta^4}{24}\right).$$

That bound is already what we want. The Lean form glues `setIntegral_mono_on` on the point-wise bound to `gaussian_tail_bound_rescaled_Ioi`, with integrability of the quartic side from the seabed’s `quartic_integrable_pow` and of the Gaussian comparator from Mathlib’s `integrable_exp_neg_mul_sq`.

The *tail-symmetry lemma* `quartic_integral_Iio_neg_eq_integral_Ioi` mirrors the seabed’s `integral_Iio_neg_eq_integral_Ioi` for the Gaussian, using `integral_comp_neg_Iic` together with the evenness identity $(-w)^4 = w^4$.

The *splitting identity* `quartic_partition_split` writes $\mathbb{R}$ as `Iic a ∪ Ioi a` (`intervalIntegral.integral_Iic_add`) and then `Iic a` as `Iio (-a) ∪ Icc (-a, a)` (`setIntegral_union`, with a hand-written disjointness witness and a six-line `ext/rcases lt_or_ge` union proof). Combined with the tail symmetry, this gives

$$\int_{\mathbb{R}} e^{-tL} = \int_{[-a,a]} e^{-tL} + 2 \int_{(a,\infty)} e^{-tL}.$$

The *headline* then unfolds in eight lines: the full-line integral equals `quartic_partition` (via `quartic_moment_even` at $n = 0$, after a small `Nat.cast_zero / mul_zero / zero_add` cleanup); the absolute-value bound on the difference equals $2 \cdot \int_{(a,\infty)}$, which by `quartic_tail_Ioi` doubled is $(24/(ta^3)) \cdot \exp(-ta^4/24)$.

The total weight of the proof is concentrated in the `setIntegral_mono_on` step (where the integrability bookkeeping costs more than the mathematical bound itself) and the union-and-disjointness glue for `setIntegral_union`. Both are mechanical once the strategy is right.

4 Roadblocks and resolutions

The deliberation surfaced two issues before any Lean was written, and the formalisation surfaced two more.

Loose exponent in the original draft. Claude’s first version of candidate A had the remainder bound $|r_a(t)| \leq C_a \cdot \exp(-ta^4/48)$, derived from a square-completion comparison $w^4/24 - w^2 \geq -6/t$. The factor 48 is correct but loose by a factor of 2 in the exponent and folds the t into an unstructured constant C_a . The numerical sanity check immediately showed the bound was $\sim 100\times$ slack at $a = 1, t = 100$, and GPT-5.5 Pro independently flagged the cleaner alternative:

GPT-5.5 Pro: A *cleaner and sharper* bound is available: $|r_a(t)| = 2 \int_a^\infty e^{-tw^4/24} dw \leq \frac{24}{ta^3} e^{-ta^4/24}$. This comes from the simple comparison $w^4 \geq a^2 w^2$ for $w \geq a$, reducing the quartic tail to the already-formalised Gaussian tail.

That same paragraph turned into the proof strategy. Calibration: the numerical check sanity-checked both the leading constant and the remainder shape, and the deliberation step caught the tightenable exponent before any Lean compilation. Two reviewer cycles saved.

Disagreement on candidate scope. Both parties backed A, but GPT also flagged a Lipschitz-near-zero middle-ground candidate (essentially candidate C with weakened hypothesis on φ) as worth considering. We deferred: the rule “one ingredient at a time” wins over “cleanest single statement that combines two ingredients.” A future tide can pick up the test-function side from this base.

Lean parser strict on `| placement`. The first version of the headline statement was written `|(( ...) - leading) | ≤ ...` with a space before the closing `|`. Lean’s parser failed at column 67 with “expected no space before”. The fix was to drop the redundant outer parens: `| ( ...) - leading | ≤ ....` The lesson is that the absolute-value notation is more sensitive to surrounding whitespace than other delimiter-like operators in Lean; in doubt, keep the inside terse.

Concurrent-agent staging race (recurrence). The repository hosting this Tide step had two other Tide branches running concurrently (`tide/2d-pure-quartic` and `tide/sextic-moments`). `Laplace.lean` was rewritten by the parallel agents twice during the run, each time stripping or reordering imports to match what was on *their* branch’s HEAD. The fix in this run was to (a) stash the working tree, (b) `git checkout main -- Laplace.lean` to a known baseline, (c) re-apply only the new `Laplace.OneD.QuarticBoundedPrior` import line and the `Laplace.TwoD.PureQuartic` import already landed on the chained-base, then (d) `git checkout -B` to retarget the branch label onto the prior tide’s tip before committing.

This is the same staging race called out in the 2D pure-quartic retrospective. The structural fix is independent of either tide: concurrent agents need filesystem isolation via `git worktree`. That’s repeated under *Lessons* below.

Beyond these four, no proof-side roadblocks. All four sub-theorems typechecked clean within ten minutes once the strategy was pinned.

5 What was Mathlib, what was new

Mathlib provided. `integrable_exp_neg_mul_sq` for the comparator Gaussian’s integrability. `setIntegral_mono_on` for monotonicity of the set integral under a pointwise inequality. `intervalIntegral.integral` and `setIntegral_union` for the splitting identity. `integral_comp_neg_Iic` for the tail symmetry. `Real.exp_le_exp`, `mul_self_le_mul_self`, `pow_pos`, and the standard `ring / field.simp / linarith` closers.

The seabed provided. The leading constant *is* the existing `Laplace.OneD.quartic_partition` — no new closed-form computation needed. The Gaussian tail `gaussian_tail_bound_rescaled_Ioi` in `Laplace.OneD.TailBound` is the input to which the quartic tail reduces; it was written for the harmonic case (used in `Localisation.lean`) and is reused here unchanged. Integrability of polynomials against the quartic Boltzmann factor came from `Laplace.OneD.quartic_integrable_pow` (also unchanged).

What was new in this tide. The four theorems in `Laplace/OneD/QuarticBoundedPrior.lean`: `quartic_tail_Ioi`, `quartic_integral_Iio_neg_eq_integral_Ioi`, `quartic_partition_split`, `quartic_partition`. None of the four theorems introduces a new mathematical technique; they wire existing seabed and Mathlib pieces in a new arrangement. The honest line-count breakdown is roughly: 5 lines of new mathematics (the comparison $w^4 \geq a^2 w^2$ and its consequence), 30 lines of arithmetic manipulation (rewriting $b = ta^2/12$, computing $\frac{1}{b \cdot a} = 12/(ta^3)$, etc.), and 200 lines of integrability and measurability bookkeeping plus splitting/union proofs. That ratio is the dominant cost on a small bounded-prior tide step in this style.

6 Lessons

- *Numerical sanity check pays for itself before writing Lean.* The exponent-constant tightening in the present run came purely from the combination of `scipy.integrate.quad` numbers and a five-minute GPT consult; it cost nothing to find and cost a factor-of-two reduction in the final bound. Worth doing on every tide whose closed form is numerical-friendly. The discipline already lives in the Tide skill; record this run as another data point that confirms it.
- *Concurrent Tide agents need filesystem isolation.* `Laplace.lean` is again the contention point; the situation is identical to the one called out in the 2D pure-quartic retrospective. Until `git worktree` support lands in the Tide skill's preflight, expect to spend a few minutes per concurrent run on the `git checkout main -- Laplace.lean → re-add imports → git checkout -B dance`. Documenting the dance here should at least make the pattern recognisable on first hit in a future run.
- *The substitution-and-Gaussian-tail template is reusable.* The exact same proof pattern (*point-wise comparison $w^{2k} \geq a^{2k-2} \cdot w^2$, then `gaussian_tail_bound_rescaled_Ioi`*) will give a one-line generalisation of `quartic_tail_Ioi` to `monomial_tail_Ioi` for $L = w^{2k}/(2k)!$. GPT-5.5 Pro flagged this in the deliberation step ("If you generalise, generalise the tail lemma first, not the full bounded-prior asymptotic theorem"); worth acting on the next time the grammar-precursor tide is taken further.
- *Lean parser is whitespace-sensitive on `|·|`.* For the seabed's `CLAUDE.md`: when writing absolute-value statements, prefer `|expr| ≤ ...` with no extra outer parens between `expr` and the closing `|`. Lean's tokeniser treats `) |` as ambiguous in some positions.

7 Follow-ups

- *Bounded-prior quartic with continuous test function* (candidate C from this deliberation). Now the next-step grammar precursor: bounded prior + tail bound infrastructure is in place; what remains is the localisation argument for φ near zero plus the `Asymptotics.IsLittleO` packaging. Estimated 400–600 lines.
- *Bounded-prior monomial generalisation* $\left| \int_{-a}^a e^{-tw^{2k}/(2k)!} dw - \frac{1}{2}((2k)!/t)^{1/(2k)}\Gamma(1/(2k)) \right| \leq C_k(a) \exp(-c_k t a^{2k})$. The tail-bound proof transfers verbatim, using $w^{2k} \geq a^{2k-2} w^2$ for $w \geq a$. The hard part is the `Real.rpow` bookkeeping that comes with the generalised exponent.
- *Bounded-prior harmonic partition* (candidate B). Now trivial given `partitionFunction_harmonic` and the splitting idiom from this tide. ~50–100 lines if anyone wants the literal Gaussian version on the record.
- *Worktree-isolated Tide runs.* Repeated from the 2D pure-quartic retrospective. The recurrence here makes the case stronger.
- *Promote the splitting/union idiom to `lean-common`.* The hand-written `Iio (-a) ∪ Icc (-a, a) = Iic a` disjointness + union proof is a self-contained, generic statement that will recur on every bounded-prior tide. Six lines is small enough to duplicate once but it should not duplicate three times.

Acknowledgements

GPT-5.5 Pro contributed the deliberation-stage advisory at Step 2 and the strategic move that became the final proof strategy: the `quartic_tail_Ioi` sublemma via the comparison $w^4 \geq a^2 w^2$. The full consult is preserved at `projects/grammar/tide-log/gpt55_tide_grammar_v1.md`. The tide log itself, with candidate drafts, vote, numerical check, and result, is at `projects/grammar/tide-log/2026-05-02-`